

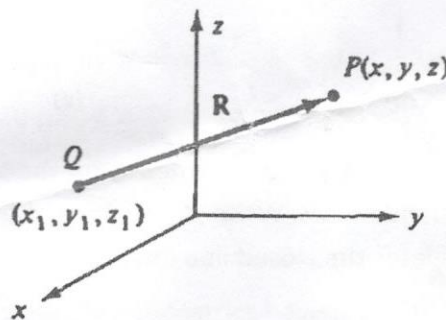


Course:	Electromagnetic Theory	Course Code:	EE305
Program:	BS (Electrical Engineering)	Semester:	Fall 2016
Duration:	180 Minutes	Total Marks:	100
Paper Date:	22-Dec-16	Weightage:	50%
Section:	ALL	Page(s):	4
Exam:	FINAL	Instructor:	Mohsin Yousuf

- Instruction/Notes:**
- (i) Attempt all questions with proper labelling of respective parts on the answer sheet.
 - (ii) Hints and instructions are in italic at the end of a question.
 - (iii) Your answer should be to the point and in the required format.
 - (iv) Marks will be deducted for not distinguishing between scalars and vectors.
 - (v) Try not to spend more than 10 min on 5 marks questions.

Q # 1. [5+5+5+5 = 20 marks]

- (a) Find the expression for the electric field E at $P(x, y, z)$ due to a point charge $Q(x_1, y_1, z_1)$ as shown in the figure below.



- (b) What will be the change in the expression of E if the point charge Q is moved to origin? Is there any symmetry in E ? If not, then how to see the symmetry in E ?
- (c) A uniform charge is distributed along an infinite straight line with constant density ρ_l [C/m]. Develop an expression for E at a general point $P(x, y, z)$ with the help of a suitable diagram.

$$\int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

- (d) Two identical uniform sheet charges with $\rho_s = 100$ nC/m² are located in free space at $z = \pm 2$ cm. Find the force per unit area each sheet exerts on the other? *Draw the diagram first.*

Q # 2. [15+5 = 20 marks]

- (a) Given the flux density $D = \frac{16}{r} \cos(2\theta) a_\theta$ [C/m²], use **two** different methods to find the total charge within the region $1 < r < 2$ m, $1 < \theta < 2$ rad, $1 < \phi < 2$ rad. *Also draw diagram.*
- (b) Calculate the total electric flux leaving the cubical surface formed by the six planes $x, y, z = \pm 7$ if the charge distribution is a uniform surface charge of $0.1 \mu\text{C}/\text{m}^2$ on the plane $y = 3x$. *Draw the diagram first.*

ρ_s

$$Q = 1.17 \times 10^{-4}$$

Q#3. [10+4+6 = 20 marks]

$$H = \frac{NI}{d}$$

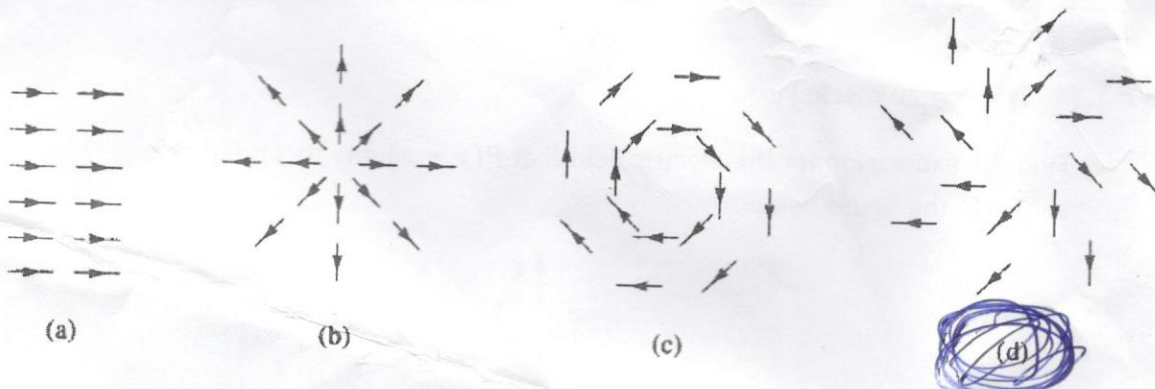
$$H = \frac{NI}{2\pi\phi}$$

torsion

(a) A vector field is uniquely characterized by its **divergence** and curl. Neither the divergence nor the curl of a vector field is sufficient to completely **describe** the field. A vector field A is said to be **solenoidal (or divergenceless)** if $\nabla \cdot A = 0$. A vector field A is said to be **irrotational (or potential)** if $\nabla \times A = 0$. Four vector fields with vanishing or nonvanishing divergence or curl are shown below.

(i) Write the physical interpretation of divergence and curl of a vector field.

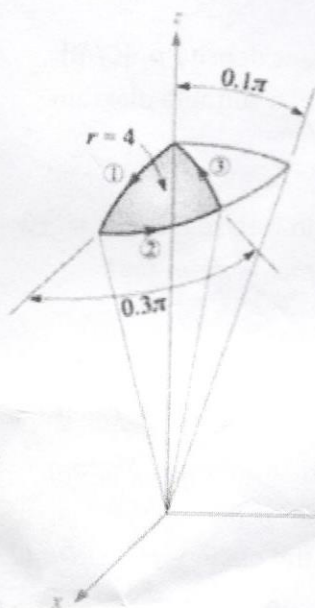
(ii) Characterize each of the following vector field in terms of divergence and curl. For e.g. if a vector field is solenoidal but not irrotational rather rotational, answer is $\nabla \cdot A = 0$ and $\nabla \times A \neq 0$. Make a table for your answer.



(b) State Biot-Savart Law for direct current in a filament with the help of a suitable diagram; clearly mentioning source and the field points. Write expressions of the law for K and J .

(c) The Stokes' theorem that relates the closed line integral to the surface integral is an identity that holds for any vector field.

$$\oint H \cdot dL \equiv \int_S (\nabla \times H) \cdot dS$$



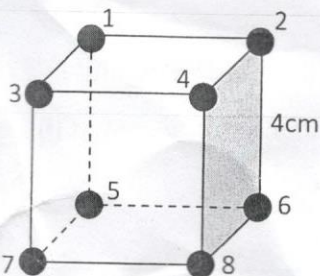
Consider a portion of the sphere as shown in the figure. The surface is specified by $r = 4$, $0 \leq \theta \leq 0.1\pi$, $0 \leq \phi \leq 0.3\pi$, and the closed path forming its perimeter is composed of three circular arcs. We are given the field $H = 6r \sin \phi \vec{a}_r + 18r \sin \theta \cos \phi \vec{a}_\phi$ (A/m). Evaluate each side of Stokes' theorem.

$$\nabla \times H = \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \vec{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial(r H_\phi)}{\partial r} \right) \vec{a}_\theta + \frac{1}{r} \left(\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \vec{a}_\phi$$

22.6A

Q #4. [15+5 = 20 marks]

- (a) Eight 2 nC point charges are located in free space at the corners of a cube 4 cm on a side as shown below. Find the total potential energy stored using the relationship,



$$W_E = \frac{1}{2} \sum_{n=1}^N Q_n V_n$$

where V_1 is the combined potential due to all charges except for the charge at 1.

- (b) A ninth charge of 2 nC is installed at the center of the cube. Again find the total stored energy.

This problem involves careful consideration of the geometry especially the distances between the point charges and the relationship of the potential due to point charge.

Q #5. [5+4+4+7 = 20 marks]

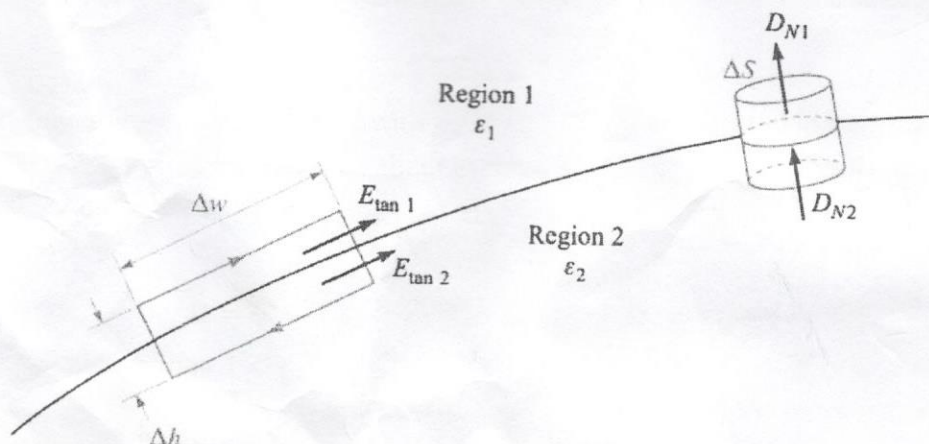
- (a) Current density is given in cylindrical coordinates as $\mathbf{J} = -10^6 z^{1.5} \mathbf{a}_z \text{ A/m}^2$ in the region $0 \leq \rho \leq 20 \text{ } \mu\text{m}$; for $\rho \geq 20 \text{ } \mu\text{m}$, $\mathbf{J} = 0$. Use appropriate symbols.

(i) Find the total current crossing the surface $z = 0.1 \text{ m}$ in the $\mathbf{a}_z$ direction.

(ii) If the charge velocity is $2 \times 10^6 \text{ m/s}$ at $z = 0.1 \text{ m}$, find ρ_v there.

(iii) If the volume charge density at $z = 0.15 \text{ m}$ is -2000 C/m^3 , find the charge velocity there.

- (b) Consider an interface between two dielectrics having permittivities ϵ_1 and ϵ_2 and occupying regions 1 and 2 as shown in the figure below. Compute the normal and the tangential components of electric field intensity $\mathbf{E}$ and electric flux density $\mathbf{D}$. Mention the laws properly.



(c) By using a refraction diagram of $\mathbf{D}$ at a dielectric interface with the condition that $\epsilon_1 > \epsilon_2$ and knowing the fact that we can find the field on one side of a boundary if we know the field on the other side, calculate the expression for magnitude and direction of electric flux density $\mathbf{D}$ using boundary conditions developed earlier in part (b).

(d) A perfectly conducting plane is located in free space at $x = 4$, and a uniform infinite line charge of 40 nC/m lies along the line $x = 6, y = 3$. Let $V = 0$ at the conducting plane. Find (i) V and (ii) $\mathbf{E}$ at $P(7, -1, 5)$. Use method of images and suitable diagram.
